Vidya Nirmaan

AI-Powered School Infrastructure Planning & Monitoring

IndiaAI Innovation Challenge — Problem Statement 5

Department of School Education, Andhra Pradesh

319	57	707	3	5
Schools	Districts	Mandals	AI Models	User Roles

March 2026 | Flutter + Supabase + FastAPI | Offline-First Mobile App

THE PROBLEM

School Infrastructure Planning is Broken

Andhra Pradesh manages 45,000+ government schools with manual, paper-based infrastructure planning. No automated fraud detection, no objective prioritization, no future-readiness.

Manual Paper-Based

Demand forms processed manually across 707 mandals

No Fraud Detection

Demand inflation goes undetected, wasting crores

Inequitable Allocation

No objective prioritization of 45,000+ schools

No Future Planning

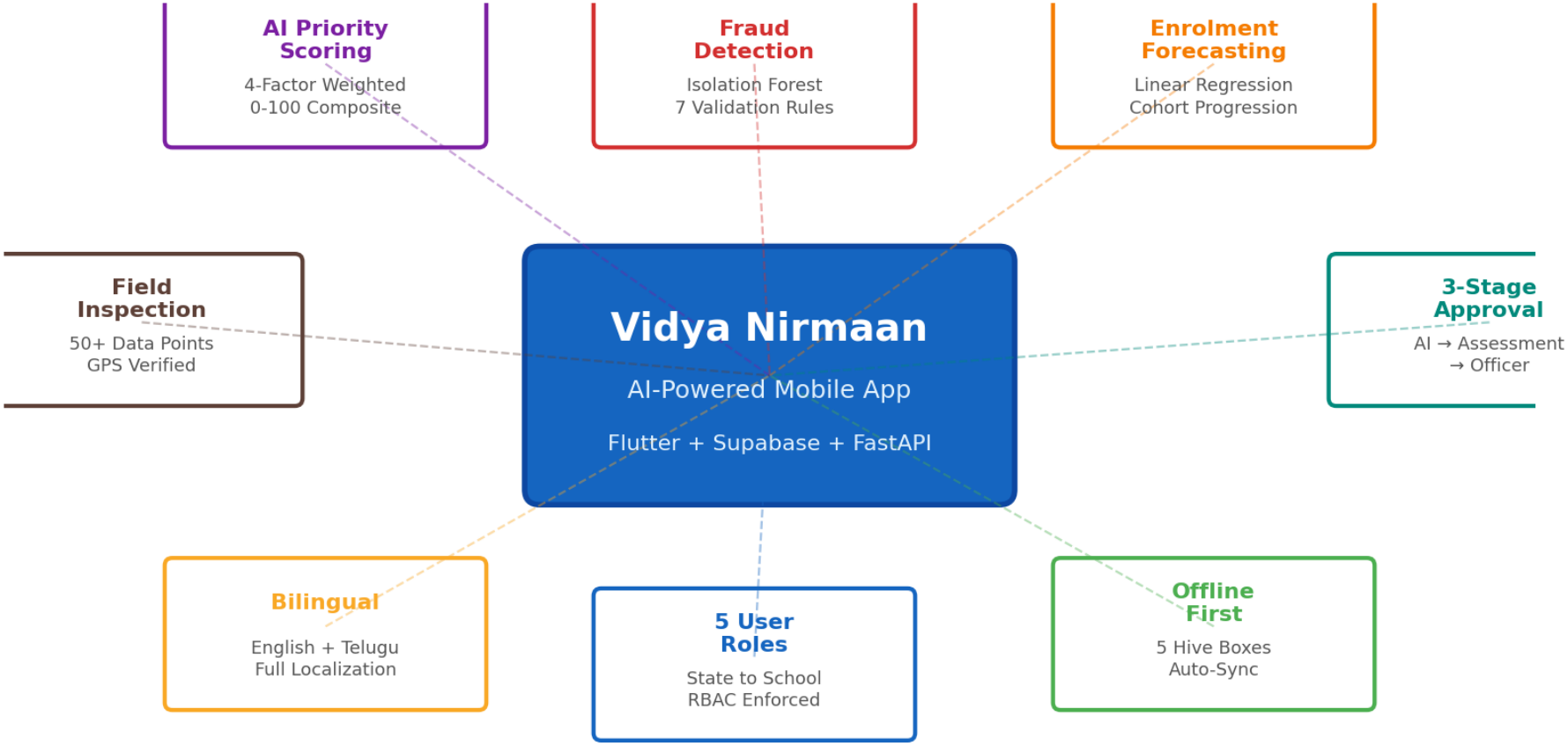
No enrolment forecasting for proactive infra planning

45,000+ Schools | 57 Districts | 707 Mandals | Andhra Pradesh | Dept. of School Education

OUR SOLUTION

Vidya Nirmaan — End-to-End AI Platform

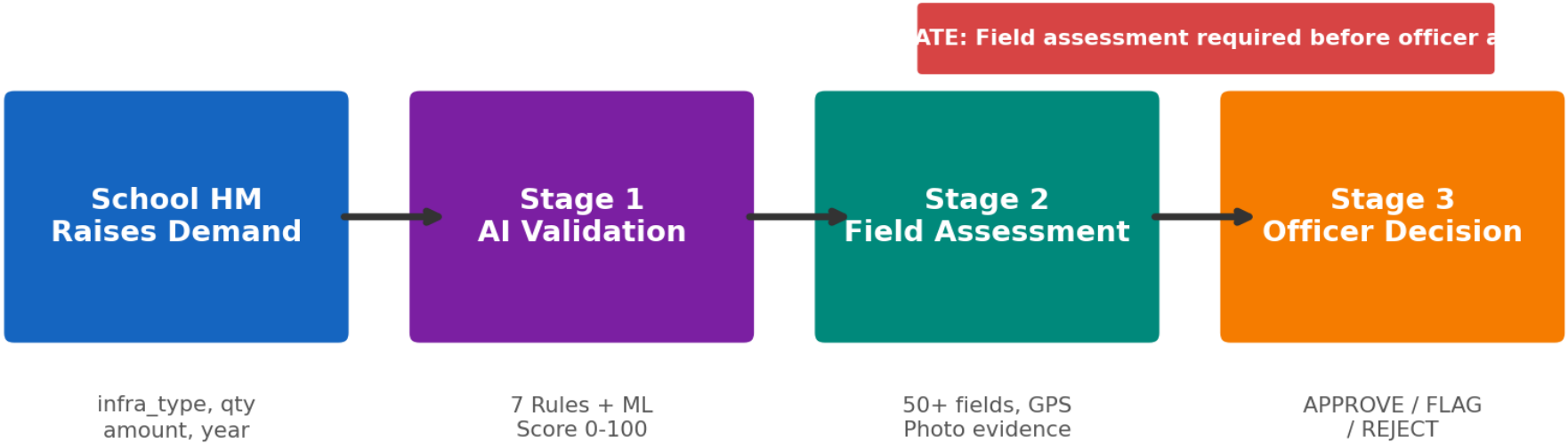
A mobile-first application that digitizes the entire infrastructure lifecycle: demand → validation → assessment → approval, powered by 3 AI/ML models.



HOW IT WORKS

3-Stage Demand Approval Pipeline

Every infrastructure demand passes through AI validation, field verification, and officer approval. A hard gate ensures no demand is approved without ground-truth inspection.



AI-First Validation

7 automated rules + Isolation Forest ML

Ground-Truth Verification

50+ inspection fields with GPS verification

Officer Oversight

Manual approve/flag/reject with notes

Full Audit Trail

Every stage timestamped and tracked

AI / ML INNOVATION

Three AI Models Working in Concert

Client-side + backend ML: priority scoring for objective allocation, Isolation Forest for fraud detection, dual-model forecasting for future planning.

Priority Scoring

School Prioritization

- Enrolment Pressure (30%)
- Infrastructure Gap (30%)
- CWSN Needs (20%)
- Accessibility (20%)

Score 0-100
CRITICAL > HIGH > MEDIUM > LOW

Demand Validation

Fraud Detection

- 7 Rule-Based Checks
- Isolation Forest ML
- Unit Cost Deviation
- Peer Comparison

APPROVED (>=80)
FLAGGED (50-79) | REJECTED (<50)

Enrolment Forecast

3-Year Prediction

- Linear Regression
- Cohort Progression
- Confidence Decay
- Backend + Client Fallback

Y1: ~93% conf
Y2: ~85% | Y3: ~76%

TECHNOLOGY

Modern, Scalable Architecture

Mobile App	Flutter 3.10+ | Riverpod 3.x | fl_chart | flutter_map | Material Design 3
State Management	Riverpod Notifier/AsyncValue | Role-Based Provider Scoping | Offline Fallback
Backend	Supabase (PostgreSQL + RLS + Auth) | FastAPI ML (Railway Cloud) | scikit-lea
Offline Layer	Hive (5 Boxes) | Auto-Sync | Queue & Retry | Graceful Degradation

Cross-Platform

Single codebase for iOS + Android

Flutter + Dart

Real-Time Backend

Supabase PostgreSQL with Row Level Security

Supabase + RLS

ML Pipeline

scikit-learn models on Railway cloud

FastAPI + Python

Offline-First

Hive local storage with auto-sync

Hive + Queue

BY THE NUMBERS

Platform at a Glance

319
Schools in Pilot Dataset

4,638
Enrolment Records Analyzed

799
Demand Plans Validated

50+
Inspection Data Points

7
Validation Rules + Isolation Forest

3
AI/ML Models Deployed

5
User Roles with RBAC Enforcement

2
Languages EN + Telugu

100%
Offline Capable Full Functionality

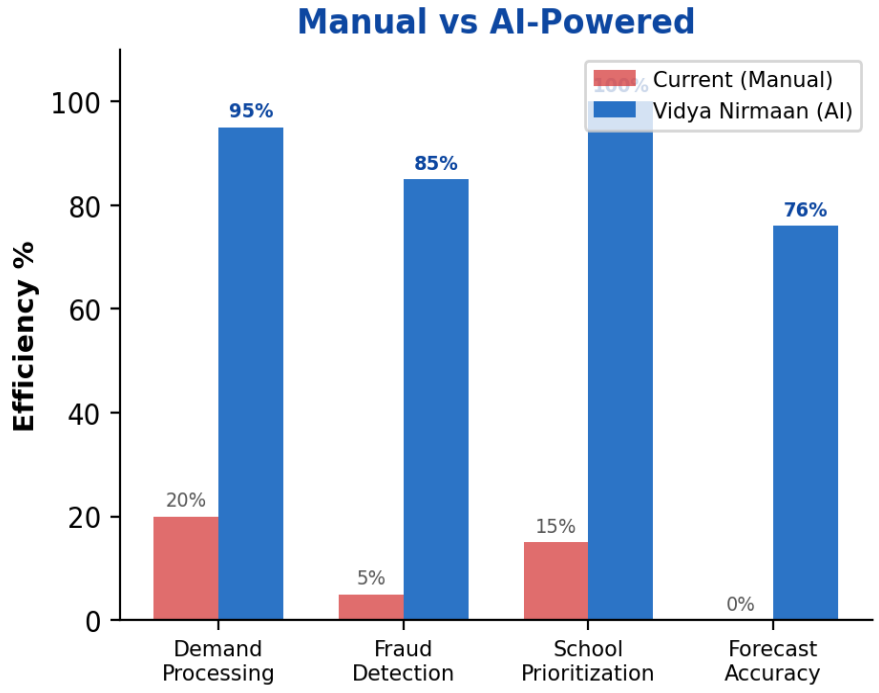
< 2s
AI Validation Response Time

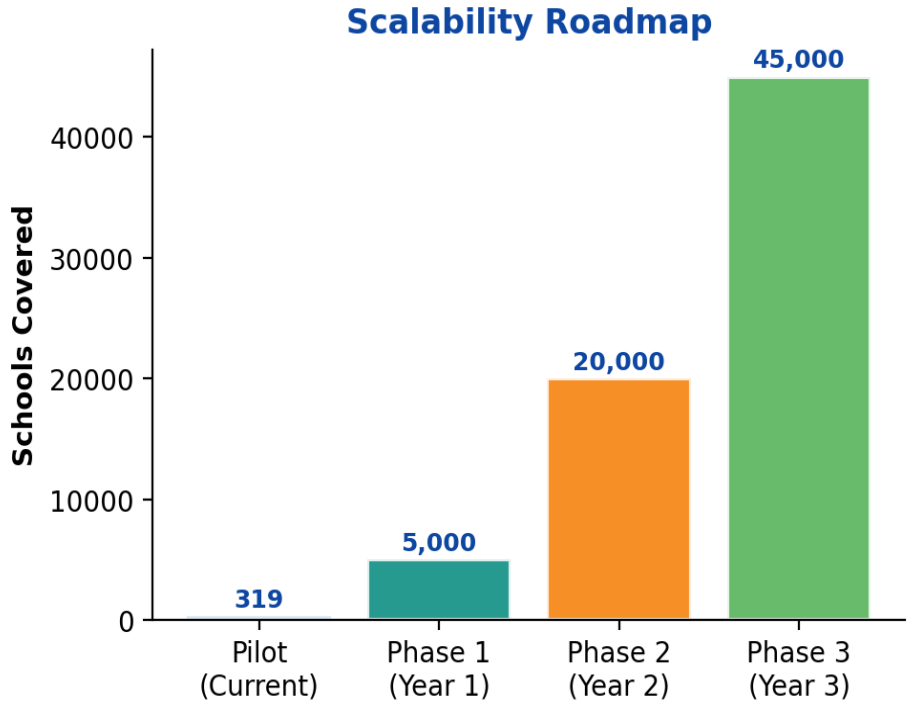
0
Manual Steps in Fraud Detection

45,000+
Schools Scalability Target

IMPACT & SCALABILITY

Transforming Infrastructure Planning





Reduced Processing Time

From weeks to seconds with AI validation

Automated Fraud Detection

Isolation Forest catches anomalies humans miss

Objective Prioritization

4-factor scoring replaces subjective decisions

Proactive Planning

3-year forecasts enable proactive investment

RESPONSIBLE AI

Ethical, Transparent, Accountable

Principle	Implementation
Transparency	All AI decisions show detailed rule-by-rule breakdown with scores. Officers see exactly why a demand was approved, flagged, or rejected.
Human Oversight	3-stage pipeline ensures human officers make final decisions. AI assists but never auto-approves without assessment + officer review.
Fairness	Objective 4-factor scoring algorithm treats all schools equally. No geographic or administrative bias in prioritization.
Privacy	No personal student data collected. Only aggregate enrolment counts. Supabase Row Level Security enforces data access boundaries.
Accountability	Full audit trail: every validation, assessment, and officer decision is timestamped with the officer name and justification notes.
Inclusivity	CWSN (Children With Special Needs) weighted at 20% in priority scoring. Bilingual interface (English + Telugu) for accessibility.
Offline Equity	Full offline-first design ensures rural schools with poor connectivity are not disadvantaged. Auto-sync when connected.

DATA GOVERNANCE

Secure, Compliant, Role-Scoped

Aspect	Approach	Details
Data Source	**Government Official Data**	UDISE+ school database, Samagra Shiksha norms, AP Dept. of Education records
Access Control	**5-Role RBAC**	State Official sees all; District Officer sees district; Block Officer/Inspector sees mandal; HM sees own school only
Database Security	**Row Level Security**	Supabase PostgreSQL RLS policies enforce data isolation per user role at the database level
Encryption	**In-Transit + At-Rest**	HTTPS for all API calls. Supabase provides AES-256 encryption at rest. JWT auth tokens for sessions
Data Minimization	**Aggregate Only**	No personal student data (names, IDs). Only aggregate enrolment counts (boys, girls, total per grade)
Offline Security	**Local-Only Cache**	Hive cache stored on device only. No cross-device data sharing. Cache cleared on logout
Audit Logging	**Full Traceability**	Every validation, assessment, and officer decision recorded with timestamp, officer name, and justification
Consent	**Government Authorization**	Data usage authorized under IndiaAI Innovation Challenge. No third-party data sharing

THE TEAM

Built by Engineers Who Ship

Role	Focus Area	Expertise
CEO	Product Vision & Architecture	End-to-end product development, Mech Engg background, AI/ML, Mobile & Cloud
COO	Operations & Strategy	Business operations, government project coordination, strategic planning
CTO	Technical Lead	Full-stack engineering, system architecture, ML pipeline development
Lead Engineer	Full-Stack Development	Full-stack engineering, 5+ years experience, frontend & backend

Organization: EdTech startup focused on AI-powered solutions for government education — school infrastructure (Vidya Nirmaan), early childhood development (ECD monitoring), and personalized learning companions for children.

Vidya Nirmaan

Transforming school infrastructure planning from manual paper-based workflows to an AI-powered, transparent, accountable, and equitable system.

3 AI Models
Priority Scoring
Validation
Forecasting

7 Validation Rules
Rule-Based +
Isolation Forest ML

5 User Roles
State to School
Role-Based Access

Offline-First
Works Without
Internet

Ready for Pilot Deployment Across Andhra Pradesh

iOS + Android | English + Telugu | Built for 45,000+ Schools